ZCAS UNIVERSITY

Computer Science Department

Human–Computer Interaction (HCI)

Group Presentation Assignement

Presentation Type: Group Presentation
Time: 5–6 minutes per group

Group 1 – Introduction to HCI

1. Explain what Human–Computer Interaction is.
2. Discuss why HCI is important in modern software systems.
3. Identify key components of HCI (user, system, interaction).
4. Give two real-world examples of HCI in daily life.

Group 2 – Usability Concepts

1. Define usability.
2. Explain the five usability attributes (learnability, efficiency, memorability, errors, satisfaction).
3. Explain why usability is important in system design.
4. Give examples of good vs poor usability.

Group 3 – User-Centered Design (UCD)

1. Explain the concept of User-Centered Design.
2. Describe the UCD process stages.
3. Explain the role of users in system design.
4. Give an example of a system designed using UCD.

Group 4 – Human Factors in HCI

1. Explain human factors in HCI.
2. Discuss cognitive factors (memory, attention, perception).
3. Explain physical factors (vision, motor skills).
4. Explain how ignoring human factors affects usability.

Group 5 – Interaction Styles

1. Explain interaction styles in HCI.
2. Describe command-line interfaces.
3. Describe menu-based and form-based interfaces.
4. Compare GUI vs CLI.

Group 6 – Graphical User Interfaces (GUI)

1. Explain what a **GUI** is.
2. Identify key GUI elements (icons, windows, menus).
3. Explain advantages of GUIs.
4. Give examples of GUI-based systems.

Group 7 – Mobile Interface Design

1. Explain challenges in mobile interface design.

2. Discuss screen size and touch interaction issues.
3. Explain usability guidelines for mobile apps.
4. Evaluate a mobile app interface of your choice.

Group 8 – Accessibility in HCI

1. Explain **accessibility** in HCI.
2. Identify users with special needs (visual, hearing, motor).
3. Discuss accessibility design principles.
4. Give examples of accessible system features.

Group 9 – Usability Evaluation Methods

1. Explain why usability evaluation is important.
2. Describe **heuristic evaluation**.
3. Describe **usability testing**.
4. Compare evaluation methods.

Group 10 – Nielsen's Usability Heuristics

1. Explain what usability heuristics are.
2. Describe any **four of Nielsen's heuristics**.
3. Explain how heuristics improve interface design.
4. Apply heuristics to evaluate a sample interface.

Group 11 – User Interface Design Principles

1. Explain UI design principles.

2. Discuss consistency, feedback, and simplicity.



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3. Explain error prevention and recovery.
4. Give examples of good UI design.

Group 12 – Prototyping in HCI

1. Explain what prototyping is.
2. Describe low-fidelity vs high-fidelity prototypes.
3. Explain benefits of prototyping.
4. Give examples of prototyping tools.

Group 13 – Emotional & Aesthetic Design

1. Explain emotional design in HCI.
2. Discuss the role of colors, layout, and visuals.
3. Explain how aesthetics affect user satisfaction.
4. Give examples of emotionally engaging interfaces.

Group 14 – Error Handling & User Feedback

1. Explain the importance of user feedback.

2. Discuss error messages and notifications.
3. Explain how systems should handle user errors.
4. Give examples of good and bad error messages.

Group 15 – HCI in Web Design

1. Explain HCI principles in web design.
2. Discuss navigation and information structure.
3. Explain responsive design.
4. Evaluate a website based on HCI principles.

Group 16 – HCI and Emerging Technologies

1. Explain HCI's role in emerging technologies.
2. Discuss HCI in AI systems.
3. Discuss HCI in VR/AR systems.
4. Identify usability challenges in new technologies.

Group 17 – Ethics in HCI

1. Explain ethical issues in HCI.
2. Discuss privacy and data protection.
3. Explain dark patterns in UI design.
4. Explain ethical responsibility of designers.

Group 18 – Future Trends in HCI

1. Explain future trends in HCI.
2. Discuss voice-based interfaces.
3. Discuss gesture-based interaction.
4. Predict how HCI will evolve in the next 10 years.